17-05-2025

Agenda :

Threading & processes

main.py

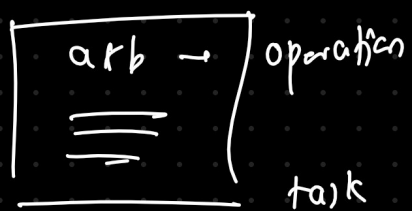
```
import _
import _
def add

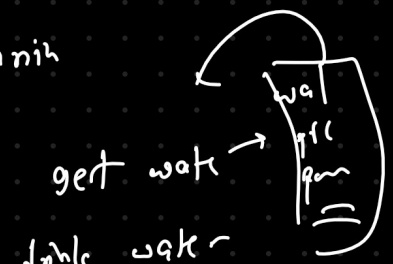

add()
```

print( ) → task

add → task

add download playing running

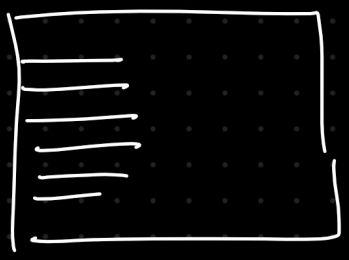




Threading

Thread : smallest unit of execution within a process

financial.py → process



process
↓
task
↓
operation

P.S. →

Top

(1) Input

(2) DOB calcu

(3) storage

(4) getter & setter

↓

Bottom

(1) process

(1) Input

(2) DOB calculator

(3) storage → 10s

(4) Input

(5) BMI calculation

(2) process

(1) Input → 10s

(2) DOB calculation → (3) storage

(4) Input ( )

(5) BMI calculator

whatsapp → clock, wallpaper, phone → stop

running in background

P.S

url: ——— .png
url: ——— .png
url: ——— .png

github → cloud
↓
image, video, code

(1) download the image → def → url → download

→ Task

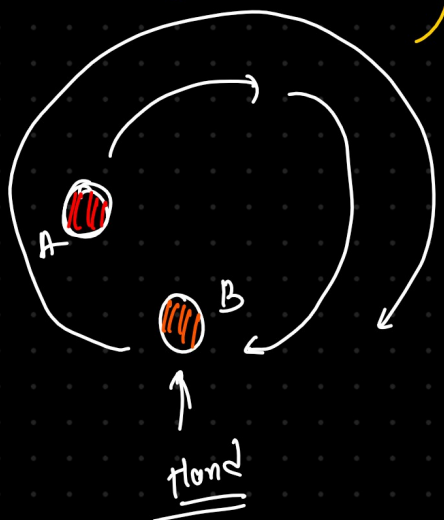
Thread

concurrently → 101. 201. | 101. 201. | 101. 201.

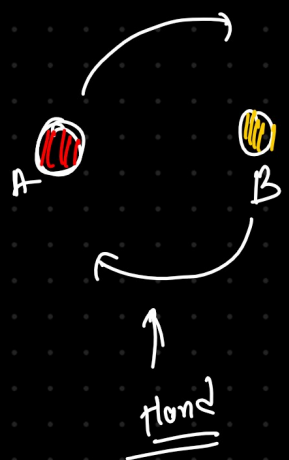
def(url) def(url) def(url)
99 100 98
100 100

(1) def(url)
(2) def(url)
(3) def(url)

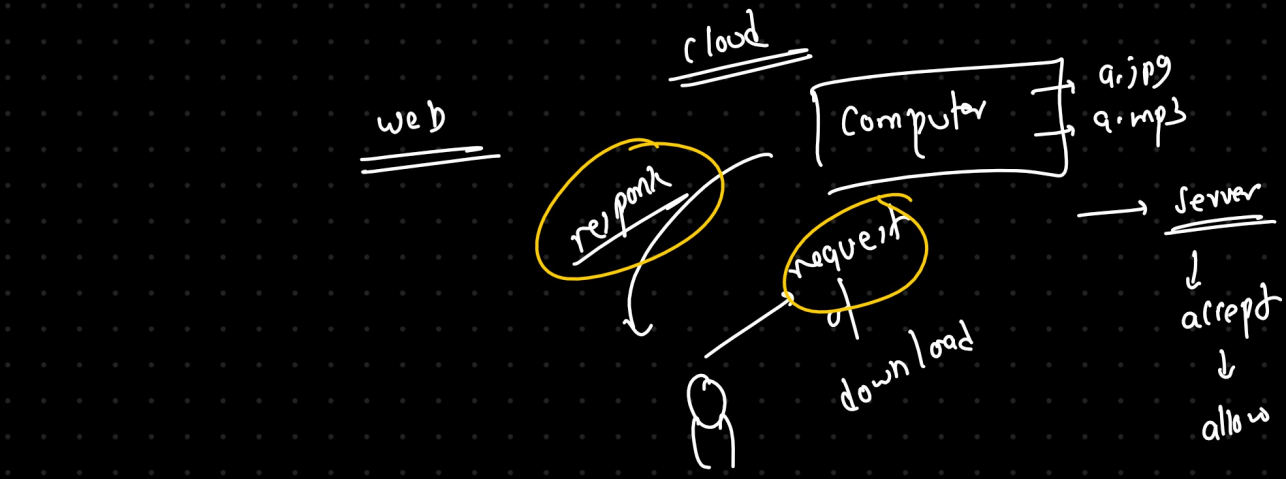
without threading

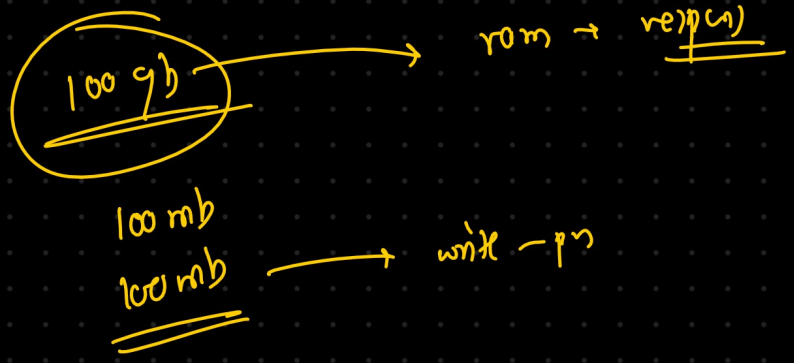


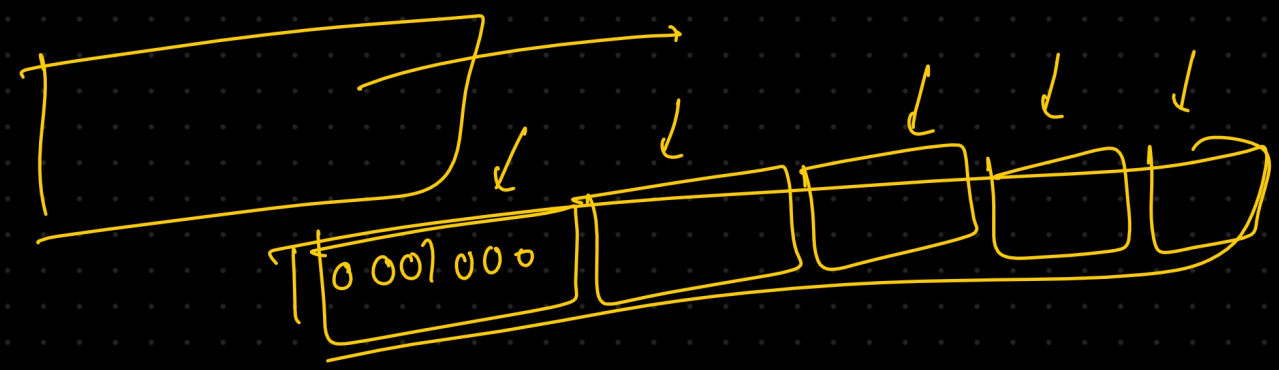
Threading

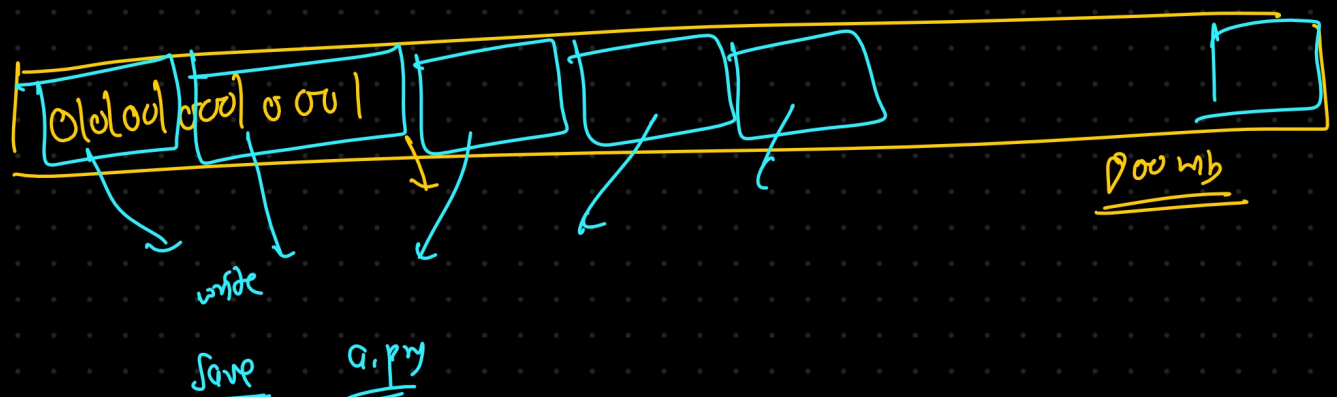


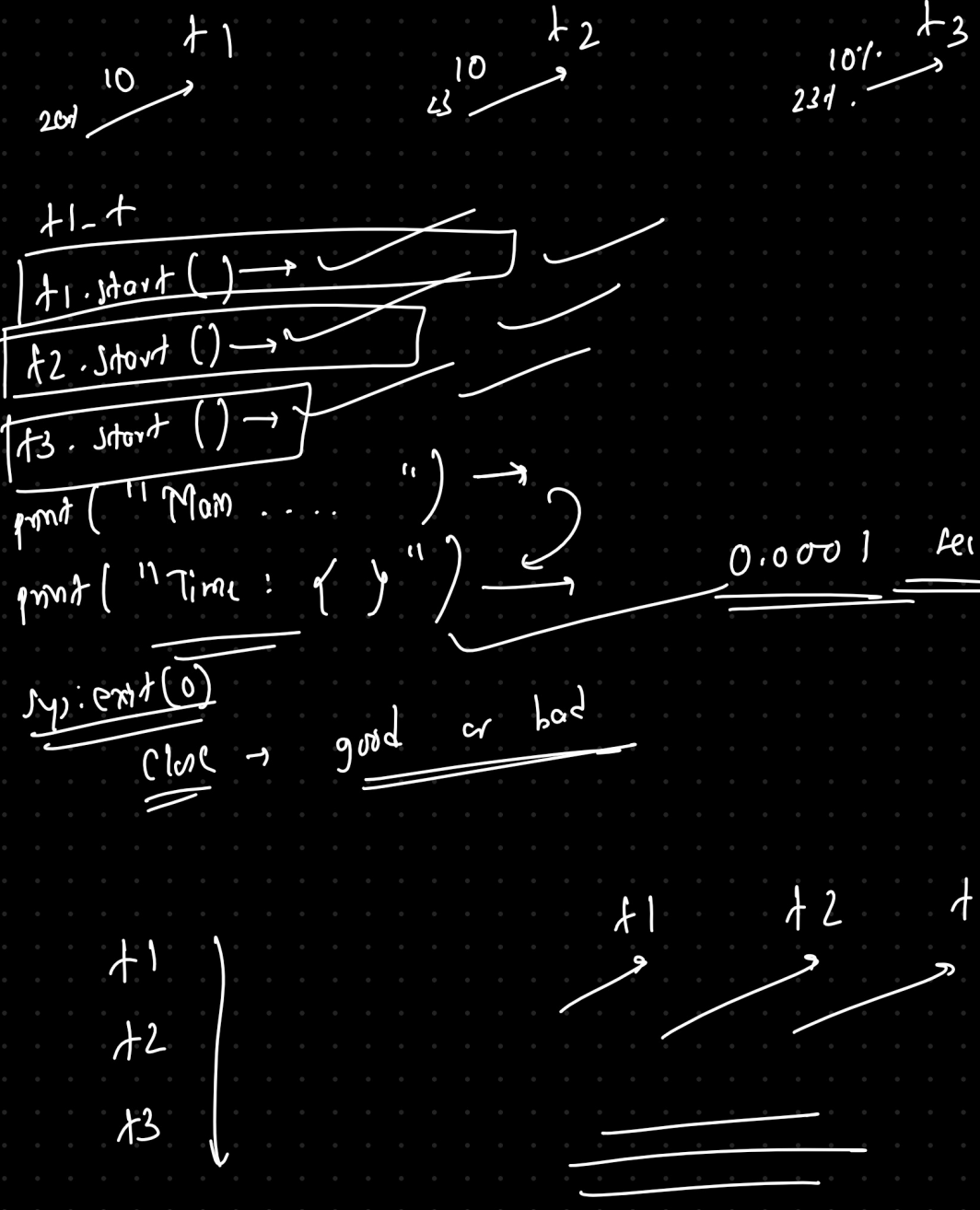
Concurrent → multiple task making progress at the same time.











pip install requests

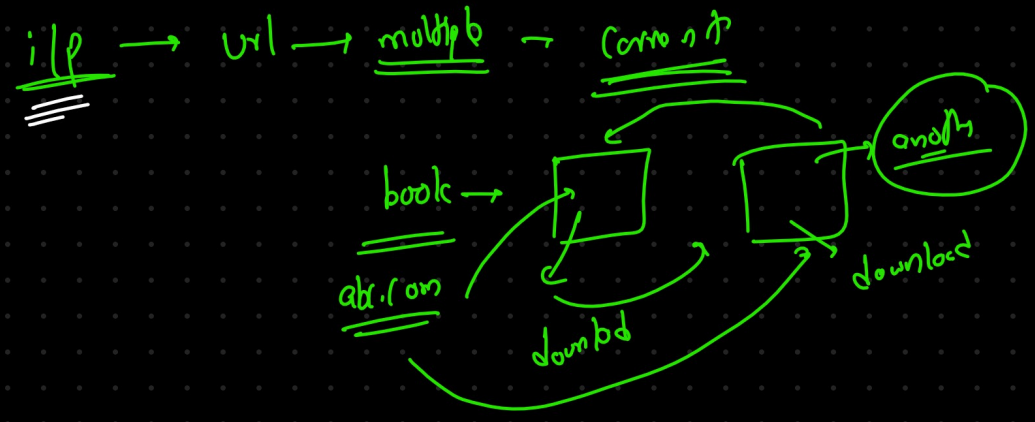
t1.start()
t2.start()
t3.start()

t3.join() → last

t1.join()
t2.join()
t3.join()
print()
print()

print()
print()



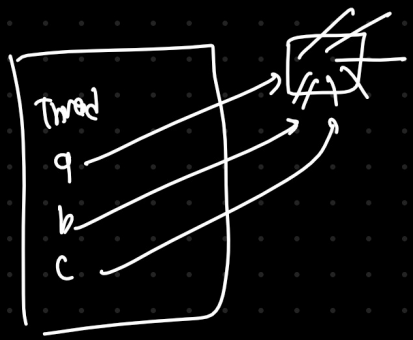


Thread →

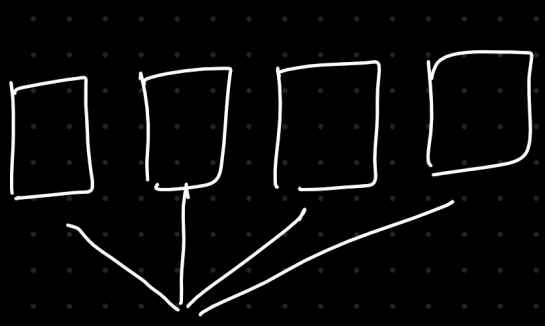
Task → One piece of work (download)

Thread → Runs one or more task

process → A program with its own memory space.



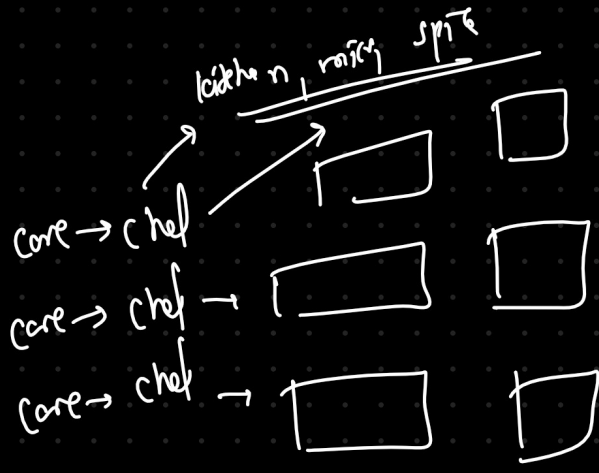
Android & iOS



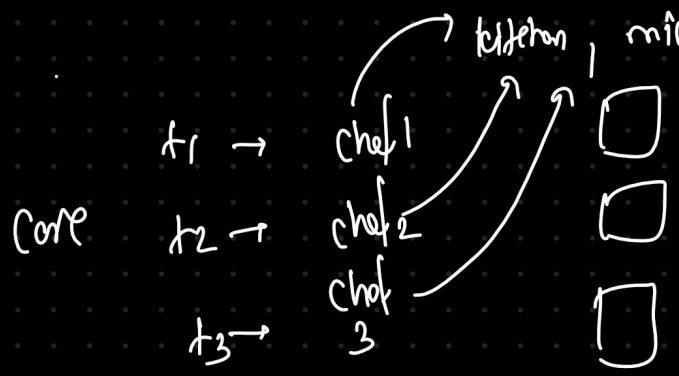
→ thread → 1 core

12 core

multiple → thread → concurrent handle task



parallel process



threading → 1

multi-threading - N

	Resource	Cores	GIL
Threading	Shared	1	One thread executes at a time
parallelism	Separate	N	Bypass GIL - Not affected

GIL → One thread execute at a time

Global Interpreter Lock

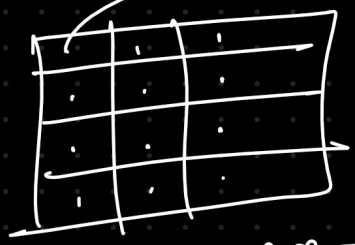
AI

CPU → 12 cores

GPU → 7168 cores

parallel →

process → independently → process → ram, memory → Threads



CPU → Core → thread

12 → Cores

12 process → separate

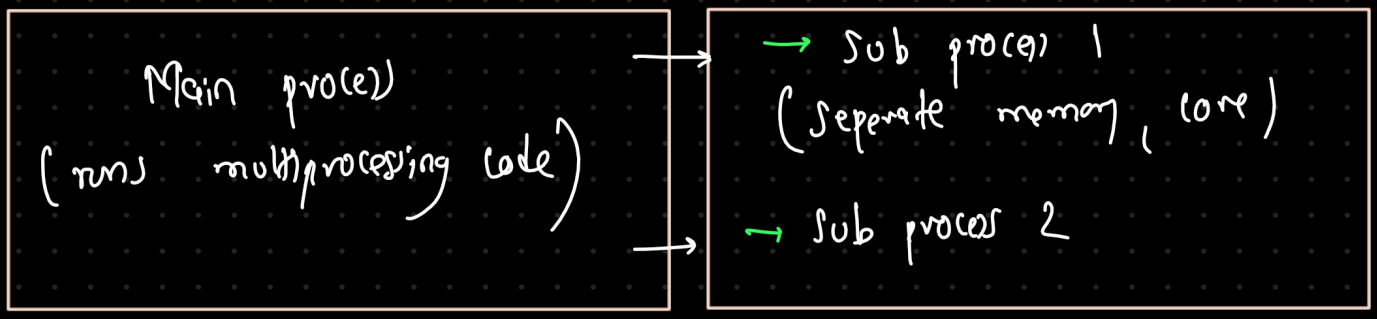
1 core → 12 process → threading → task → download

task → download

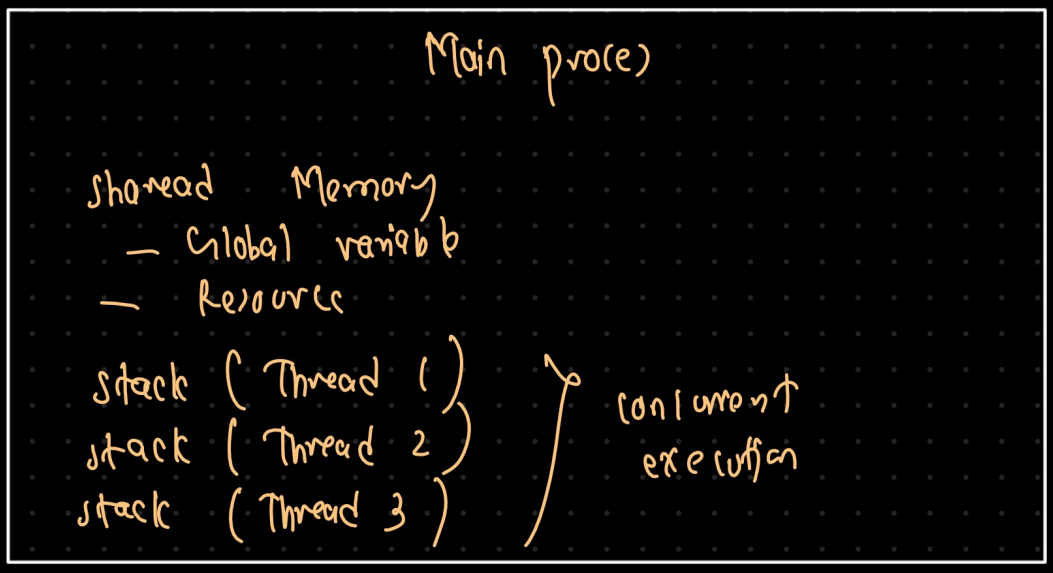
task → download

→ shared →

Multiprocess



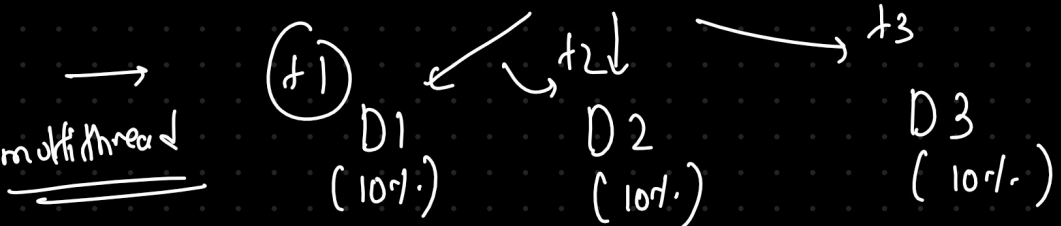
Multithreading

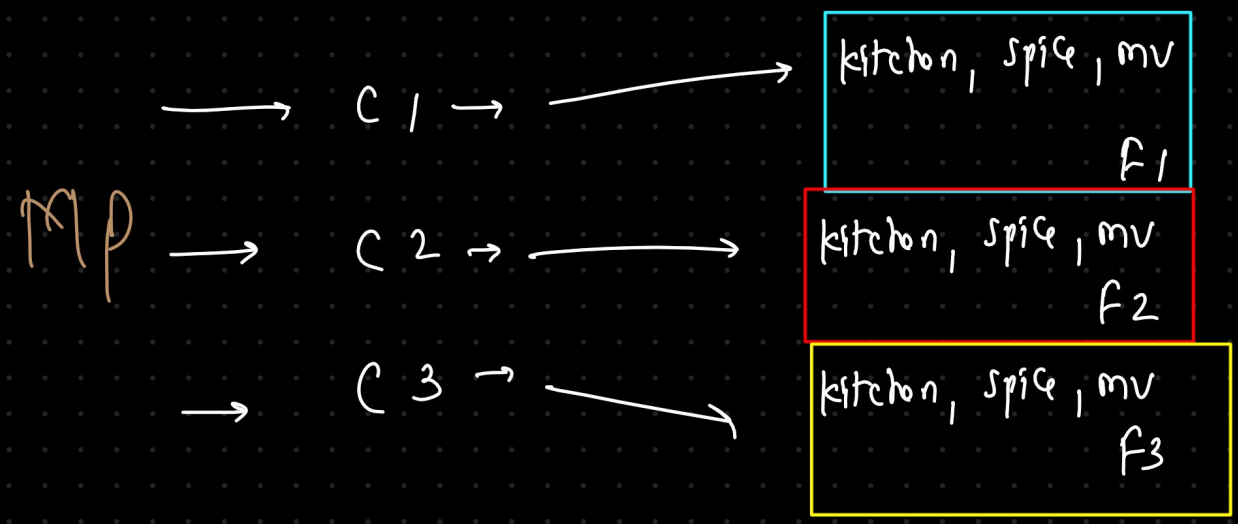


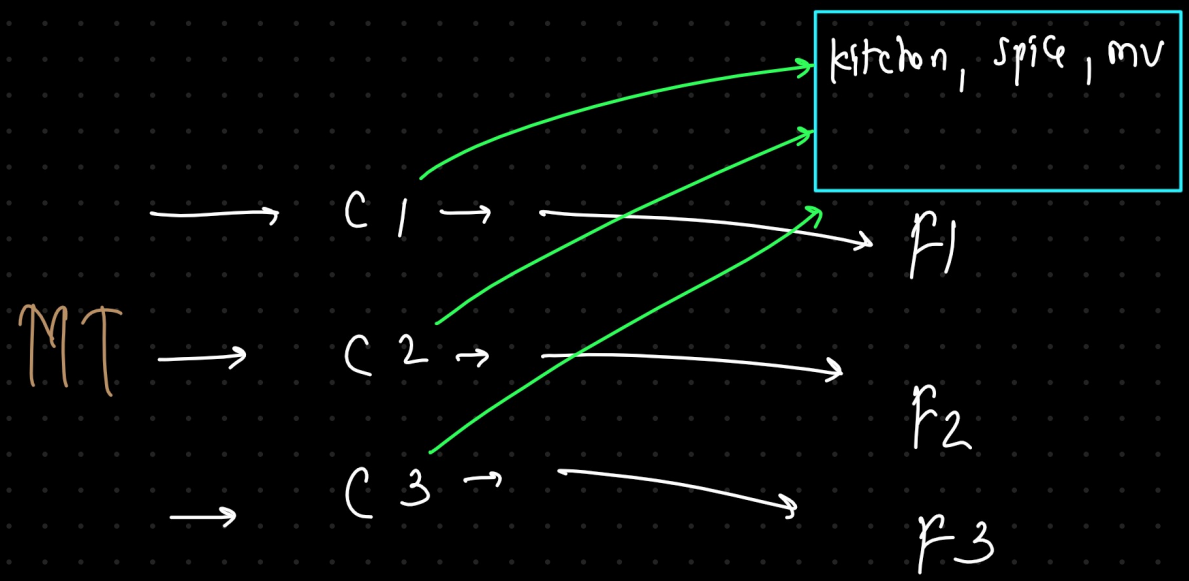
Each thread:

Has its own stack (function calls, local variable)

But accesses the same heap memory



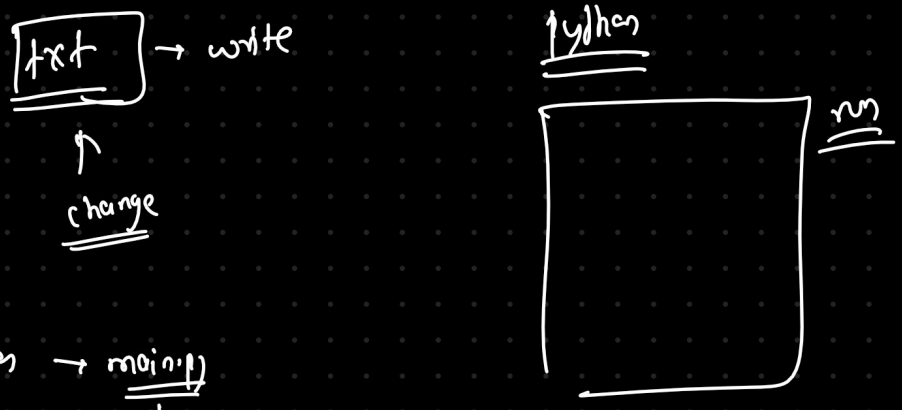


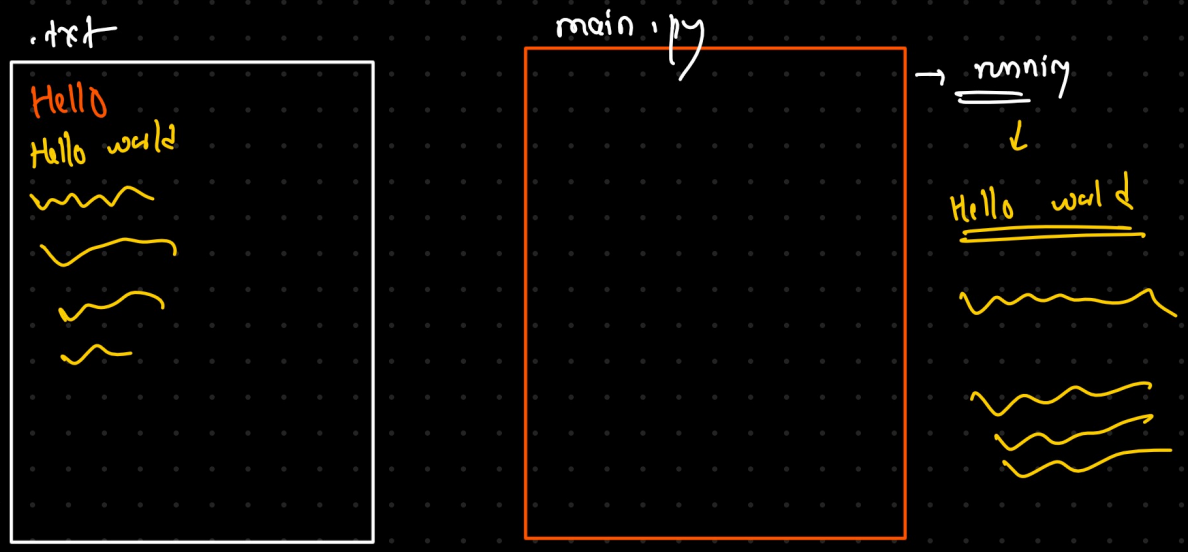


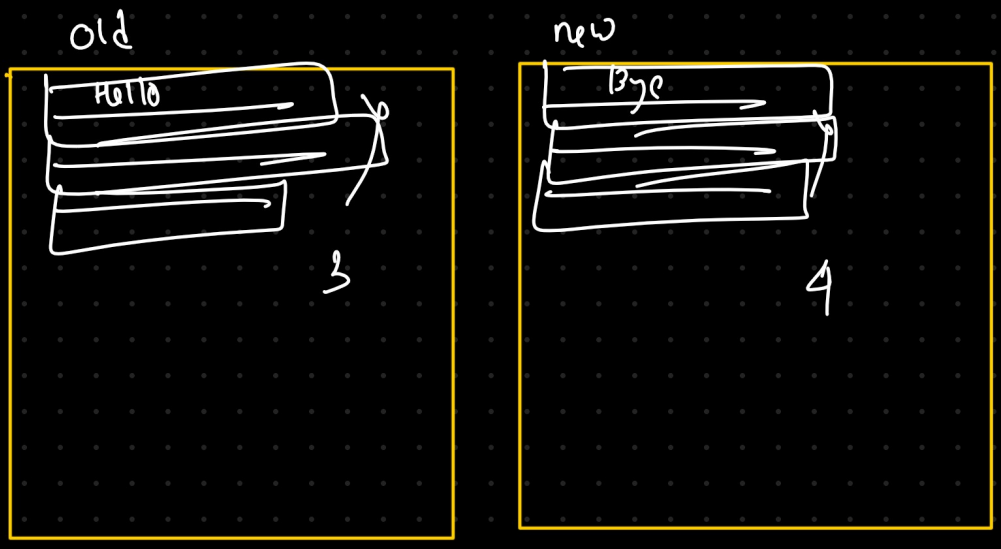
MT → Download, I/Op

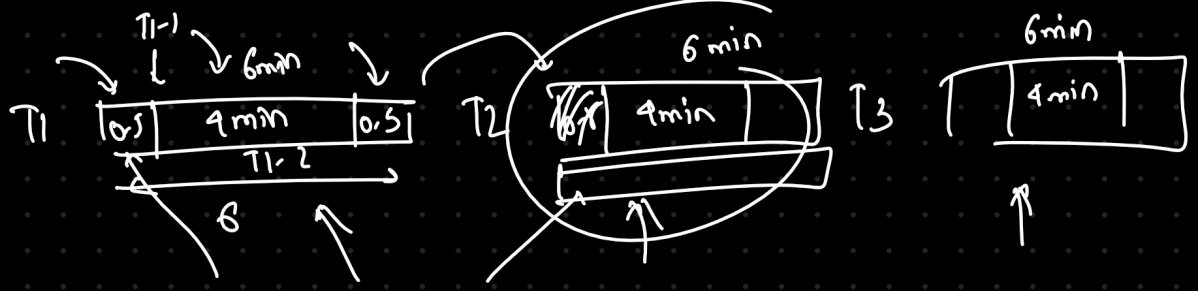
MP → Model Training, weather prediction, search analysis

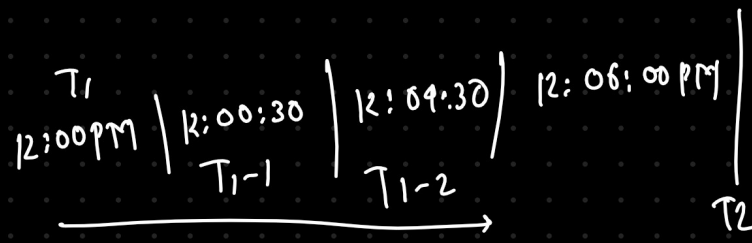
P.I :

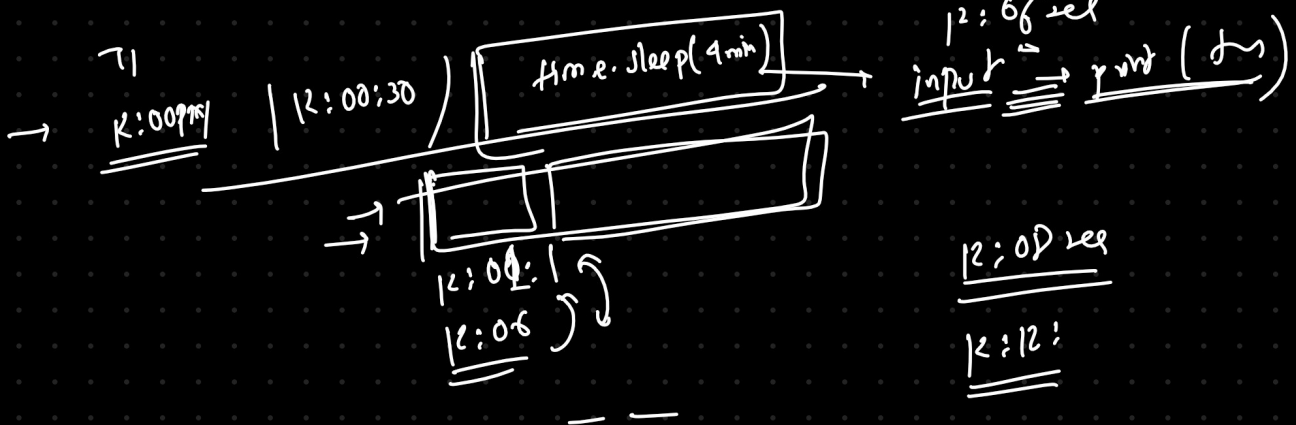

.txt
write
change
Python
run
Hello → java → main.py
world
world

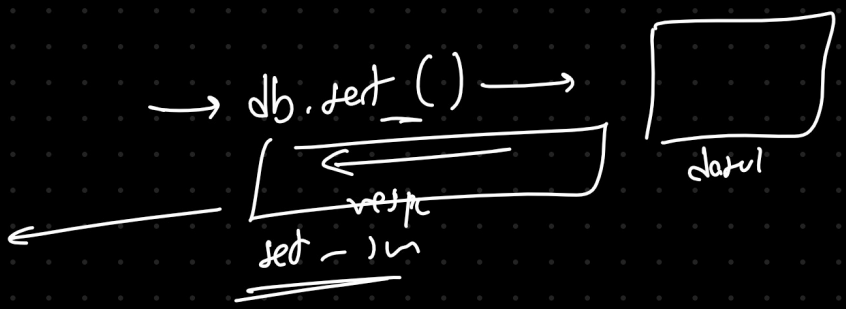

.txt
Hello
Hello world
main.py
running
Hello world


old
Hello
3
new
Bye
4


T1-1
6min
T1
0.5
4min
0.5
T1-2
6
T2
6 min
4min
T3
6min
4min


T1
12:00PM
12:00:30
T1-1
12:04:30
T1-2
12:06:00PM
T2


T1
12:00PM
12:00:30
time.sleep(4min)
12:06 rec
input
print (data)
12:04:1
12:06
12:04 rec
12:12:


db.set_()
data1
resp
set - ok